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SECTION 02

Assignment 1

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1.0 Knowledge Representation

1.1 Character Revision Frequency

Defining repetition frequency of a certain Mandarin character to appear as a question based on the mastery of the particular character.

Rule:

```
IF user_answer_correctly = false
    OR (error_rate > 0.6 AND lessons_since_last_revision >= short_interval)
    OR (error_rate > 0.3 AND error_rate <= 0.6
        AND lessons_since_last_revision >= medium_interval)
    OR (error_rate <= 0.3 AND lessons_since_last_revision >= long_interval)
THEN show_character_question = true
```

Explanations:

The system determines the appearance frequency of questions determined by user performance and time interval since the last exercise done on the particular Mandarin character. There are several conditions of determining whether a question repeat for practice:

- Character that user got wrong in one learning session is repeated immediately
- Character with high error rate (> 0.6) is repeated in a short period of time
- Character with medium error rate (> 0.3) is repeated in a medium period of time
- Character with error rate ≤ 0.3 is repeated occasionally over a long period of time

This rule ensures that revision frequency of learned Mandarin characters is adjusted accordingly based on error rate and spacing to strengthen memory efficiently.

Predicate Logic/First-Order Logic:

```
 $\forall u \forall c \forall e \forall n ($   
     $User(u) \wedge Character(c) \wedge ErrorRate(u, c, e) \wedge LessonsSinceRevision(u, c, n) \wedge$   
     $(\neg UserAnsweredCorrectly(u, c)$   
     $\vee (e > 0.6 \wedge n \geq ShortInterval)$   
     $\vee (0.3 < e \leq 0.6 \wedge n \geq MediumInterval)$   
     $\vee (e \leq 0.3 \wedge n \geq LongInterval))$   
     $\rightarrow ShowForRevision(u, c)$   
 $)$ 
```

For any user u and character c , the character is shown for revision if **any** of the following conditions are true:

1. The user answered it incorrectly in the last session.
2. The user has a high error rate and the short interval has passed.
3. The user has a medium error rate and the medium interval has passed.
4. The user has a low error rate and the long interval has passed.

1.2 Handwriting Recognition

Predicting correctness of Mandarin character writing using stroke order, direction and similarity rules.

Rule:

*IF user_stroke_order = correct_order
 AND user_stroke_direction = correct_direction
 AND user_stroke_similarity >= threshold
THEN handwriting_recognition = correct*

Explanation:

The “Rydit: Learn Mandarin” system reviews user’s handwriting Chinese characters in real time to check whether each character is written correctly. The system examines 3 main features of the handwriting:

1. **Stroke Order:** Rydit checks whether the user writes the strokes in the same order as the standard character.
2. **Stroke Direction:** It analyzes the direction of each stroke to determine it matches the correct writing method.
3. **Stroke Shape and Accuracy:** The system compares the user’s stroke shapes to the standard version. If the shapes are close enough within set tolerance levels, the stroke is considered correct.

When all 3 conditions are met, Rydit predicts that the user has written the character correctly. The app then provides immediate positive feedback, helping the user build confidence and develop consistent writing habits. These predictions are based on clear rules and data analysis, ensuring users receive reliable guidance throughout their Mandarin handwriting practice.

Predicate Logic/First-Order Logic:

$\forall u \forall c \forall i (CorrectOrder(u, c, i) \wedge CorrectDirection(u, c, i) \wedge Similar(u, c, i) \rightarrow PredictCorrect(u, c, i))$

For every user u , character c and stroke i , if the user u writes stroke i of character c in the correct order and the stroke direction is correct and the stroke is similar with standard version, then the system predicts the stroke is correct.

1.3 New Word Introduction

New characters shall be introduced from time to time to ensure progression. The condition to introduce a new word and what word to be introduced has to be determined.

Condition to introduce a new word:

Rule:

IF user_accuracy_in_last_5_attempts $\geq$ 0.9 THEN introduce_new_word

Explanation:

The system introduces a new character to the user once they have demonstrated mastery of a current character by achieving a 90% accuracy rate or higher in their last 5 attempts. This indicates that the user has a low error rate and is ready to move on to a new character for further learning.

Predicate Logic/First-Order Logic:

$\forall u \forall c \{User(u) \wedge Character(c) \rightarrow$
 $[\forall i (i \in 1, 2, 3, 4, 5) \wedge Attempt(u, c, i, accuracy) \rightarrow (accuracy \geq 90)$
 $\rightarrow NewWordIntroduced(u, c)]\}$

For every user u and character c , if the user has achieved 90% accuracy ($accuracy \geq 90$) on their last 5 attempts i , then the system will introduce a new word (character c) to user u .

Word to be introduced:

Rule:

IF character_not_mastered THEN next_word_to_be_introduced

Explanation:

The system maintains a list of characters that are sorted by their difficulty. The difficulty is based on factors such as relevance, usage frequency, and radical complexity. Characters that have not been mastered by the user and are available for learning will be selected and introduced to the user in the next learning session.

Predicate Logic/First-Order Logic:

$\forall u \{User(u) \rightarrow \exists c [Character(c) \wedge \neg Mastered(u, c) \wedge AvailableCharacter(c)]$
 $\rightarrow NextWord(u, c)\}$

For every user u , if there exists a character c that has not been mastered ($\neg Mastered(u, c)$) and is available for learning ($AvailableCharacter(c)$), it will be selected as the next word to introduce.

1.4 Motivation & Reward Mechanism

Reward to engage the users in an immersive learning experience by providing the rewards and points that motivate consistent learning and reinforcement.

Rule:

*IF user_progress_increases OR consistent_practice = true
OR difficult_question_solved = true OR high_accuracy_in_revision = true
THEN reward_points_increases AND positive_feedback_triggered*

Explanation:

The system encourages the learners by giving reward points and providing positive feedback whenever they perform good learning behaviours. For example, this includes making progress while studying, practising consistently, correctly answering difficult questions, or completing a set of problems with high accuracy. As the reward points increase, the system provides motivating feedback to the user. Various achievements can be unlocked, and badges are awarded when users reach certain targets, keeping them engaged and immersed in the learning process.

Predicate Logic/First-Order Logic:

$\forall u \forall c \text{ Progress}(u, c) \vee \text{ConsistenPractice}(u) \vee \text{SolvedDifficult}(u, c) \vee \text{HighAccuracy}(u, c) \\ \rightarrow \text{IncreaseRewardPoints}(u) \wedge \text{TriggerPositiveFeedback}(u)$

For every user u and character c , if the user shows progress, practises consistently, solves a difficult question, or achieves high accuracy during studying and revision, then the system will increase the user's reward points and provide positive feedback to the respective user.

1.5 Adaptive AI Feedback

Adaptive AI Feedback provides personalised guidance to learners by analysing their writing performance in real time. The system evaluates stroke correctness, accuracy patterns and user progress to generate appropriate feedback. This ensures that each learner receives helpful hints, corrections or motivational messages tailored to their current needs, improving the learning experience and making Mandarin writing practice more effective.

Rule:

IF stroke_error_detected=true

*OR repeated_mistakes_on_character=true OR low_accuracy_in_recent_attempts=true
OR performance_improving=true*

THEN provide_personalised_feedback AND adjust_guidance_level

Explanation:

The system delivers different types of personalised feedback depending on the user's writing behaviour. When the user makes a stroke error, such as writing a stroke in the wrong direction or order, the AI gives immediate micro-feedback to correct the mistake. If the user repeatedly struggles with the same character, the system offers more detailed corrective guidance and suggests targeted practice.

When recent accuracy drops, the system provides supportive hints and simplified explanations to help the learner recover. However, if the user shows improvement, such as making fewer mistakes or achieving more accurate strokes, the system triggers encouraging feedback to reinforce progress.

This personalised feedback mechanism adapts to the learner's performance level, ensuring they stay motivated while receiving the right type of guidance at the right time.

Predicate Logic/First-Order Logic:

$$\forall u \forall c [(StrokeError(u,c) \vee RepeatedMistake(u,c) \vee LowAccuracy(u,c) \vee Improving(u,c)) \rightarrow (ProvideFeedback(u,c) \wedge AdjustGuidance(u))]$$

For every user u and character c , if the user makes a stroke error, repeatedly struggles with the same character, shows low accuracy, or demonstrates improvement, the system will provide personalised feedback and adjust the guidance level accordingly.

2.0 Knowledge Representation Involved to Achieve Goals

The major goal of **Rydit: Learn Mandarin** is to provide learners with an interactive, adaptive and efficient method to learn Chinese characters. In this AI system, knowledge representation involves the use of **logical representations** which are propositional logic and predicate logic to capture user performance, handwriting accuracy, revision history and engagement. These representations allow the AI to make informed decisions about which characters to include in lesson, when to provide feedback and how to introduce new content, ensuring optimal learning outcomes for each individual user.

The system uses **character revision frequency rules** to determine how often each character should reappear for practice. Characters that are recently incorrect, have high error rates or are partially mastered are shown more frequently, while well-mastered characters are reviewed once in a while only. The **Leitner-inspired spaced repetition** strategy is consistently applied to strengthen memory retention without overwhelming the learner.

Handwriting recognition is another critical knowledge representation component. The system models each stroke of a character and predicts whether the user has written the stroke correctly, then provides real-time feedback on stroke order, direction, and overall accuracy. This allows the system to guide learners step-by-step in writing characters correctly, building skill and confidence.

New word introduction is represented with rules that track user mastery over current characters. By analyzing the user's accuracy, the system introduces new characters once sufficient mastery is demonstrated. Only characters not yet mastered and available for learning are selected as the next word. This knowledge representation ensures progressive and personalized learning.

Additionally, Rydit incorporates **motivation and reward mechanisms** to sustain engagement. User actions, progress and achievements are captured as predicates, allowing the AI to assign points, badges or unlock challenges. These rules ensure that learners are consistently motivated and remain immersed in the learning process.

By integrating **adaptive AI feedback, dynamic spaced repetition, handwriting recognition, and motivational rules**, the Rydit system represents a **knowledge-driven and personalised approach** to Mandarin learning. The predicates and rules capture relationships and conditions essential for decision-making, allowing the AI to deliver content, feedback and progression in a way that strengthens retention, improves writing accuracy and maintains learner engagement.